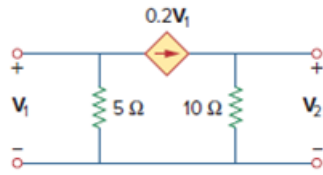


Problem

Obtain the admittance parameter equivalent circuit of the two-port in Fig. 19.82.

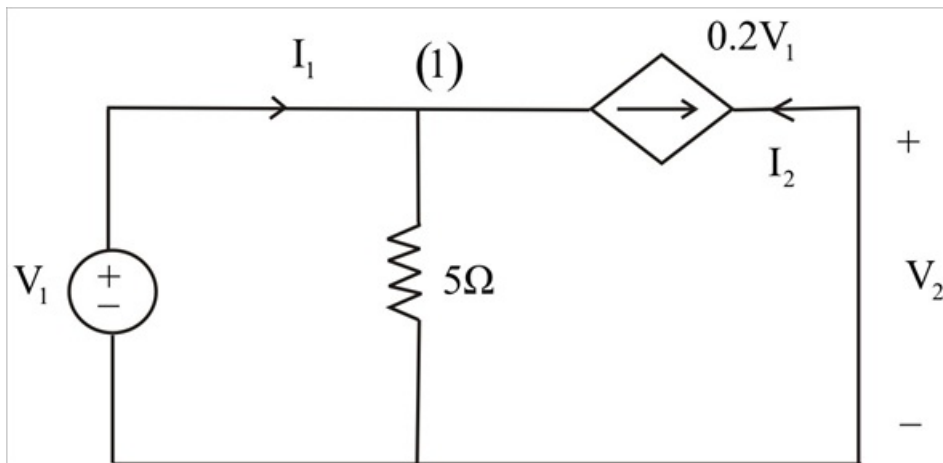
Figure 19.82



Step-by-step solution

Step 1 of 5

For $V_2 = 0$ the circuit becomes



Step 2 of 5

$$\text{At Node 1, } I_1 = \frac{V_1}{5} + 0.2V_1$$

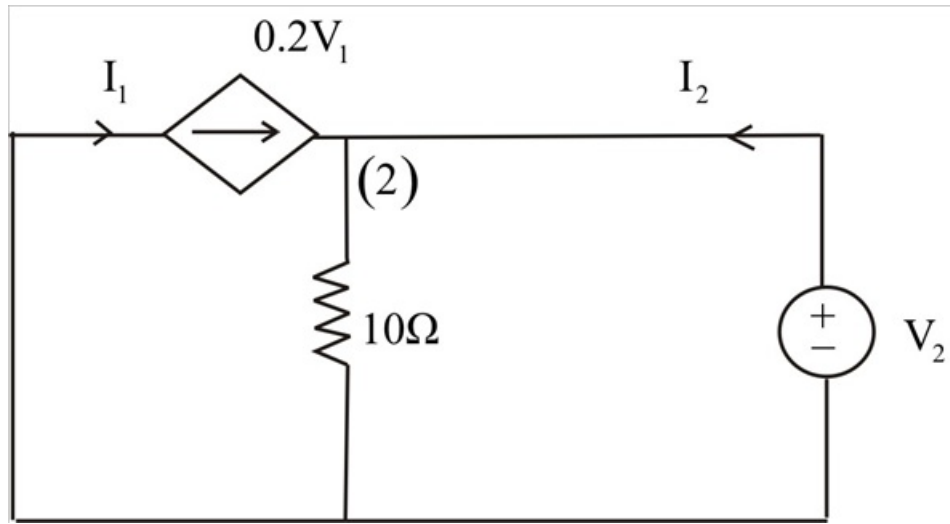
$$\frac{I_1}{V_1} = \frac{1}{5} + \frac{2}{10} = \frac{2}{5} S$$

$$Y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0} = \frac{2}{5} S$$

$$Y_{21} = \left. \frac{I_2}{V_1} \right|_{V_2=0} = \frac{-0.2V_1}{V_1} = -\frac{1}{5} S$$

For $V_1 = 0$ the circuit becomes

Step 3 of 5



Step 4 of 5

At Node 2
$$I_2 = \frac{V_2}{10} - 0.2V_1$$

$V_1 = 0$

Therefore
$$\left. \frac{I_2}{V_2} \right|_{V_1=0} = \frac{1}{10} S$$

$$Y_{12} = \left. \frac{I_1}{V_2} \right|_{V_1=0} = \frac{0.2V_1}{V_2} = 0$$

$$Y = \begin{bmatrix} \frac{2}{5} & 0 \\ -\frac{1}{5} & \frac{1}{10} \end{bmatrix}$$

Equivalent circuit is given by

Step 5 of 5

